

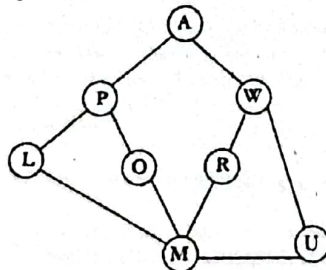
University of Dhaka
Affiliated Engineering College
Department of Computer Science and Engineering
2nd Year 1st Semester Final Examination – 2023

Course Code: CSE 2101
Time: 3 hours

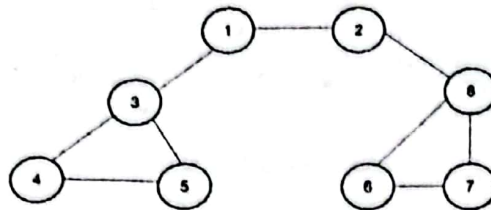
Course Title: Data Structures and Algorithms
Marks: 70

[Answer any Five Questions]

1. a) Define and classify data structure. Explain the basic operation performed on data structure. [4]
b) Briefly explain asymptotic notation. [4]
c) Using the Bubble sort algorithm find the number of comparison and number of interchanges to sort PEOPLE alphabetically. [4]
d) Write down the complexity of following algorithms: [2]
i. Merge sort ii. Binary Search
iii. Selection Sort iv. Radix sort
2. a) Consider the following Dequeue of characters where Dequeue is a circular array which is allocated six memory cell: Front=2 and Rear=4 [6]
Dequeue: _, A, C, D, _, _
Describe the Dequeue while the following operation take place
i) F is added.
ii) Two letters are deleted from the front.
iii) K, L and M are added.
b) What properties are mandatory to have for a recursive procedure? Show the recursive solution of Tower of Hanoi method for disk 4 using a schematic diagram. [5]
c) What are garbage collection, overflow and underflow? [3]
3. a) What is linked list? Explain different types of linked list with pictorial view. [5]
b) What are the benefits of using linked list over array? [3]
c) Write an algorithm to perform the following operations: [6]
i) Searching a node from a linked list
ii) Create a Circular linked list
4. a) Write the PUSH and POP algorithms. How can you implement a stack using array? [4]
b) Show the successive steps for finding 40 and 100 from the following list of elements using binary search algorithm: 11, 22, 30, 33, 40, 44, 55, 60, 66, 77, 80, 99. [5]
c) Construct a Binary Search Tree (BST) for the following sequence of numbers- 50, 70, 60, 20, 90, 10, 40, 100 [5]
5. a) How many ways a graph can be traversed? Explain BFS and give the traverse output step by step of the following graph.(starting node A) [4]



- b) Show the DFS traverse output steps of the following graph. (Starting node 1). [3]



- c) A binary tree T has 9 nodes. T inorder and preorder traversals of T yield the following sequences of nodes. Draw the diagram of the Tree: [4]

Inorder: M N O P Q R S T U

Preorder: Q N M P O S R U T

- d) Prove that a graph with n connected nodes having (n-1) links can be treated as Tree. [3]
6. a) Construct an expression tree for the expression $(a + b * c) + ((d * e + 1) * g)$. Give the outputs when you apply preorder, inorder and postorder traversals [5]
- b) Write down the characteristics of Huffman coding. What is prefix rule? Explain with example. [4]
- c) A file contains the following characters with the frequencies as shown. Find the Huffman Code for each character, prove that Huffman coding compresses the data set successfully. [5]

Characters	Frequencies
a	10
e	15
i	12
o	3
u	4
s	13
t	1

7. a) Define Heap. Write down the properties of max-heap and min-heap. [3]
- b) Construct a Max-heap from the following list of numbers: [6]
- 34, 30, 40, 22, 50, 2, 77, 55
- i) Insert 70 into the max-heap
- ii) Delete 20
- Show all the necessary steps.
- c) If set A contains 13 elements, set B contains 8 elements and the intersection of these two sets contains 5 elements, then find the number of elements in A union B. [3]
- d) Show that set $A = \{2, 5, 6\}$ and set $B = \{4, 7, 8\}$ are disjoint sets. [2]